

Introduction to Hardware Design

Shared Memory: Figures and Code Snippets

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0.1 AXI-MM queue

```
class PolyCmdHdr(DataList):
    elements = {
        "read_ind": {
            "schema": IntField.specialize(bitwidth=16, signed=False),
            "description": "Read pointer",
        },
        "write_ind": {
            "schema": IntField.specialize(bitwidth=16, signed=False),
            "description": "Write pointer",
        },
        "queue_len": {
            "schema": IntField.specialize(bitwidth=16, signed=False),
            "description": "Maximum queue length in number of entries"
        }
    }
}

void kernel(uint64_t queue_base,
            volatile uint32_t *doorbell,
            ap_uint<mem_dwidth> *mem) {
    #pragma HLS INTERFACE s_axilite port=queue_base
    #pragma HLS INTERFACE s_axilite port=doorbell
    #pragma HLS INTERFACE m_axi      port=mem
    #pragma HLS INTERFACE ap_ctrl_none port=return
    while (1) {
        // Poll doorbell
        if (*doorbell != 0) {

            // Clear doorbell
            *doorbell = 0;

            // Read queue entries from memory
            // Process all outstanding commands
            // Write completions back to memory

        }
    }
}
```